

Step	Step Description	Agents Involved	Tool Calls and Parameters	Reason
1	User requests to find all tall houses (over 18m or 7 storeys) in Exeter	human, host_agent	Message from human to host_agent: 'Find all the tall houses that are over 18m or 7 storeys tall in Exeter'	The user query is passed to the host agent to initiate the workflow.
2	Plan the solution steps	planning_agent	send_message to planning_agent with task_description: 'User wants to find all tall houses (over 18m or 7 storeys tall) in Exeter. What are the general steps to solve this query using available agents?'	The planning agent is chosen to break down the query into actionable steps, as per its system prompt.
3	Planning agent decomposes the query	planning_agent	None (internal reasoning)	The planning agent identifies the area (Exeter), the entity (tall houses), and the conditions (over 18m or 7 storeys), as per its prompt and the user query.
4	Find the area polygon for Exeter	named_area_agent	send_message to named_area with task_description: 'Find the area polygon for Exeter.'	The named_area_agent is chosen because the query is about a named area (Exeter), and the agent's prompt specifies it handles such queries.
5	Retrieve named area polygons	named_area_agent	call_os_ngd with parameters: filters=None, bbox=None, polygon_or_point=True, street_address=None, filename='exeter_polygon'	The call_os_ngd tool is used to fetch all named area polygons, as per the agent's prompt. No filters are applied to ensure all possible Exeter polygons are retrieved.
6	Filter for the main Exeter polygon	data_analysis_agent	send_message to data_analysis_agent with task_description: Filter the artifact 'exeter_polygon' to find the specific area polygon for Exeter. Save as 'filtered_exeter_polygon'.	The data_analysis_agent is chosen to filter the raw polygons to the main city polygon, as the named_area_agent's output may contain multiple entries.
7	Generate metadata for exeter_polygon	data_analysis_agent	generate_metadata_for_artifacts with artifact_names=['exeter_polygon']	Metadata is generated to understand the columns and structure of the artifact, as per the coding agent's template.
8	Filter Exeter polygon using code	coding_agent, code_executor	code_executor with code that filters for name1_text == 'Exeter', descriptiongroup in ['Settlement'], description in ['City', 'Town']	The code uses 'name1_text', 'descriptiongroup', and 'description' columns based on metadata and the need to uniquely identify the city polygon.
9	Confirm filtered Exeter polygon	data_analysis_agent	Output: Filtered Exeter polygon: 1 entry found. Artifact 'filtered_exeter_polygon' generated.	The result confirms the correct polygon for Exeter is isolated for further spatial queries.
10	Search for buildings in Exeter polygon	buildings_agent	send_message to buildings with task_description: Search for houses in the area defined by 'filtered_exeter_polygon' that are over 18m tall or have more than 7 storeys.	The buildings_agent is chosen because the query is about buildings within a specific area, as per its prompt.
11	Retrieve all buildings in Exeter area	buildings_agent	call_os_ngd with parameters: filters=None, bbox='filtered_exeter_polygon', polygon_or_point=None, street_address=None, filename='all_buildings_exeter.geojson'	The bbox parameter is set to the filtered Exeter polygon to restrict the search to the city area. No filters are applied to get all buildings for further filtering.
12	Filter for houses and tall/storey criteria	data_analysis_agent	send_message to data_analysis_agent with task_description: From 'buildings_all_buildings_exeter.geojson', filter for houses (buildinguse_addresscount_residential > 0 and buildinguse_addresscount_total = 1), then select those over 18m tall (relativeroofbase > 18) or more than 7 storeys (storeycount > 7). Save as 'houses_over18m_or_7storeys_exeter.geojson'.	The data_analysis_agent is chosen to apply complex filtering logic, as per the buildings_agent's special OS scenario for defining houses.
13	Generate metadata for buildings artifact	data_analysis_agent	generate_metadata_for_artifacts with artifact_names=['buildings_all_buildings_exeter.geojson']	Metadata is generated to confirm the presence and names of relevant columns for filtering.
14	Attempt to filter for tall houses using code	coding_agent, code_executor	code_executor with code: filter for houses (buildinguse_addresscount_residential > 0 and buildinguse_addresscount_total = 1), then filter for height_relativeroofbase_m > 18 or numberoffloors > 7	The code uses the columns specified in the special OS scenario and metadata. Several attempts are made to execute the code correctly.
15	Final code execution and result	code_executor	code_executor returns: The search returned no results and no artifacts are generated	The code is executed successfully, but no houses in Exeter meet the criteria.
16	Communicate results to user	data_analysis_agent, host_agent	Output: No houses in the Exeter buildings dataset meet the criteria. No results were found and no output file was generated.	The result is communicated back to the user, explaining the definition of a house and the applied criteria, as per the agent's prompt and special OS scenario.

Overall Summary: - Databases used: Named area database (for Exeter polygon), buildings database (for all buildings in Exeter). - Filters/code: The main Exeter polygon was filtered using 'name1_text', 'descriptiongroup', and 'description' columns. Buildings were filtered for houses using 'buildinguse_addresscount_residential > 0 and buildinguse_addresscount_total = 1', and then for height ('height_relativeroofbase_m > 18') or storeys ('numberoffloors > 7'). - Outcome: No houses in Exeter (as defined by a single residential address) are over 18m tall or have more than 7 storeys. No results or output files were generated. - Reasoning: Each agent and tool was chosen based on the system prompts and metadata. The definition of a house was strictly followed as per the special OS scenario. All filtering steps used column names and logic derived from metadata and agent/system prompts to ensure accuracy and compliance with the query requirements.